

FRC Research — Week 4 Working Notes

4 μ s Formation Extension and Transition to GS/MRR Tilt Reproduction

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1. Purpose of These Notes

These are the working research notes for Week 4 of the FRC project.

Week 3 completed the software-takeover and inherited-baseline reproduction stage. The original 2000-step WarpX formation case was parameterized, generated through Framework v0.1, executed on SCNet, and quantitatively compared against the two frozen legacy realizations.

The principal result carried into this week is

Framework v0.1 inherited-baseline regression: PASS

for the macroscopic magnetic-field and density evolution of the inherited formation case.

The purpose of Week 4 is therefore to:

1. perform one controlled extension of the validated formation case from $2\ \mu\text{s}$ to the full imposed-field rise time $t_{\text{peak}} = 4\ \mu\text{s}$;
2. characterize the additional $2 \rightarrow 4\ \mu\text{s}$ evolution without reopening the completed baseline-regression work;
3. freeze formation as a validated framework example;
4. move the main research effort to the Grad–Shafranov / Modified Rigid Rotor equilibrium and three-dimensional tilt-instability problem.

2. Handoff from Week 3

At the end of Week 3, the inherited 2000-step formation case had been reproduced using Framework v0.1.

The validated reference configuration has

$$N_t = 2000, \quad \Delta t = 1\ \text{ns}, \quad t_{\text{end}} = 2\ \mu\text{s},$$

while the imposed external-field ramp uses

$$t_{\text{peak}} = 4\ \mu\text{s}.$$

Thus, the inherited case terminates when the drive amplitude is only

$$A_{\text{time}}(2\ \mu\text{s}) = \sin\left(\frac{\pi(2\ \mu\text{s})}{2(4\ \mu\text{s})}\right) = \frac{1}{\sqrt{2}} \simeq 0.7071.$$

This motivated one final formation run extending the same validated model to

$$t = 4\ \mu\text{s},$$

where

$$A_{\text{time}} = 1.$$

The purpose of this extension is exploratory rather than regression-based: there is no frozen 4000-step legacy run against which to compare it.

3. Research Log — 2026-09-28

3.1. Decision to Extend the Formation Case to 4 μs

The 2000-step baseline was intentionally left unchanged and retained as the validated regression reference.

A separate case configuration was derived for the longer simulation:

```
configs/formation/formation_4us.toml
```

with the intended case identity

```
name = "formation_4us_ramp_complete_4000"
purpose = "formation_extension"
max_steps = 4000
```

The timestep was not changed:

$$\Delta t = 10^{-9} \text{ s},$$

so

$$t_{\text{end}} = 4000\Delta t = 4.0 \times 10^{-6} \text{ s} = 4 \mu\text{s}.$$

No physical model, geometry, particle loading, transport model, field-drive formula, or boundary condition was intentionally changed relative to the validated baseline.

The longer run therefore isolates the effect of continuing the already validated formation simulation through the remainder of the imposed-field rise.

3.2. Output Strategy

The plotfile diagnostic cadence was retained so that the full evolution could be inspected every 100 steps.

Checkpoint interval

$$N_{\text{chk}} = 2000$$

was also retained, producing checkpoints at approximately

$$t = 0, 2, 4 \mu\text{s}.$$

The checkpoint at step 2000 provides a direct connection to the endpoint of the previously validated baseline, while the final checkpoint preserves the new 4 μs state for possible later restart or analysis.

3.3. SCNet Execution

The formation extension was submitted through Framework v0.1 as Slurm job

```
123207160
```

and executed on

```
a02r2n15
```

with the same CPU layout used by the validated baseline:

$$4 \text{ MPI ranks} \times 8 \text{ OpenMP threads/rank} = 32 \text{ allocated CPU cores}.$$

The completed accounting record was

```
State      = COMPLETED
Elapsed    = 05:29:56
ExitCode   = 0:0
```

The run produced approximately

18 GB

of output, including

41

plotfile diagnostics and

3

checkpoint directories.

This is consistent with diagnostics from step 0 through step 4000 in increments of 100 and checkpoints at steps 0, 2000, and 4000.

4. Research Log — 2026-09-29

4.1. Quick Analysis Objective

The purpose of the analysis was not to repeat the full Week 3 regression.

Instead, the analysis focused on the new interval

$$2 \mu s \leq t \leq 4 \mu s,$$

while retaining the earlier evolution as context.

The main quantities inspected were

$$B_{z,\text{center}}, \quad B_{z,\text{min}}, \quad B_{z,\text{max}},$$

$$\rho_{\text{center}}, \quad \rho_{\text{max}},$$

and

$$T_{i,\text{center}}, \quad T_{i,\text{max}}.$$

Spatial $x - z$ midplane slices and radial B_z profiles were also generated at

$$t = 2, 3, 4 \mu s.$$

As in the earlier baseline audit, $B_z = 0$ is used only as a simple field-reversal indicator. It is **not** treated as a formally reconstructed magnetic separatrix.

4.2. Selected Scalar States

Table 1 summarizes representative times from the full $0 \rightarrow 4 \mu s$ evolution.

The most negative central magnetic field over the complete run occurred at

$$t \simeq 1.7 \mu s,$$

with

Table 1: Selected states of the 4 μs formation extension.

t [μs]	A_{time}	$B_{z,c}$ [T]	$B_{z,\text{min}}$ [T]	ρ_c	ρ_{max}
0.0	0.000	-0.0800	-0.0800	8.01	8.07
1.0	0.383	-0.0801	-0.1414	7.93	21.64
1.7	0.619	-0.3947	-0.4831	36.24	44.15
2.0	0.707	-0.1240	-0.2507	12.67	50.82
3.0	0.924	-0.1464	-0.1982	12.86	68.29
4.0	1.000	-0.3093	-0.3103	24.03	80.19

$$B_{z,\text{center}} \simeq -0.395 \text{ T}.$$

The most negative chamber field also occurred at approximately the same time:

$$B_{z,\text{min}} \simeq -0.483 \text{ T}.$$

This confirms that the strongest early magnetic compression does **not** occur at the end of the imposed-field ramp.

4.3. Non-Monotonic Magnetic Response

The imposed drive rises monotonically from

$$A_{\text{time}} = 0$$

to

$$A_{\text{time}} = 1$$

at 4 μs .

The plasma response is not monotonic.

The central field evolves approximately as

$$-0.080 \text{ T} \rightarrow -0.395 \text{ T} \rightarrow -0.124 \text{ T} \rightarrow -0.309 \text{ T},$$

at

$$t = 0, 1.7, 2.0, 4.0 \mu\text{s},$$

respectively.

The current working interpretation is therefore

$$\boxed{\text{strong compression} \rightarrow \text{relaxation} \rightarrow \text{renewed compression}}$$

rather than a quasi-static monotonic response that simply follows the imposed field ramp.

The words “relaxation” and “renewed compression” are presently descriptive. A more rigorous dynamical interpretation would require additional energy, flux, force-balance, and topology diagnostics.

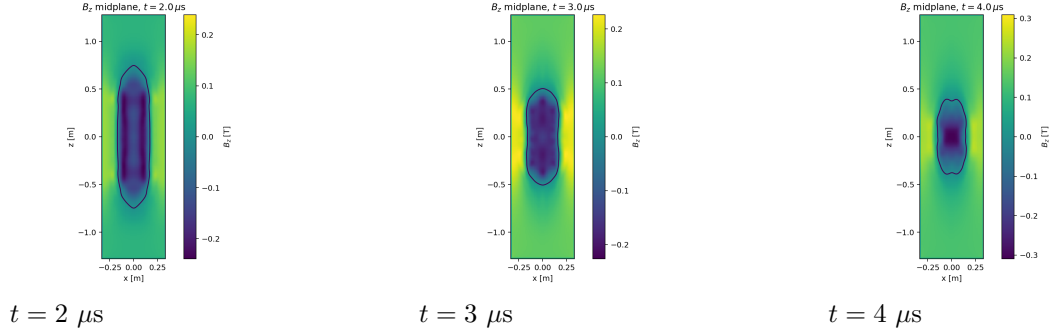


Figure 1: Midplane $B_z(x, z)$ evolution during the second half of the imposed-field rise. The black $B_z = 0$ contour, when present, is used only as a field-reversal proxy and not as a formal magnetic separatrix.

4.4. Magnetic Structure from 2 to 4 μs

Figure 1 shows the midplane axial-field structure at the validated baseline endpoint and at the two additional times analyzed during the extension.

The simulation retains regions of both negative and positive axial field at the end of the field ramp.

At $t = 4 \mu\text{s}$,

$$B_{z,\text{center}} \simeq -0.309 \text{ T},$$

while

$$B_{z,\text{max}} \simeq +0.245 \text{ T}.$$

Therefore, field reversal remains present when

$$A_{\text{time}} = 1.$$

This result demonstrates persistence of a reversed-field structure through the complete imposed-field rise, but it is not by itself sufficient to establish the exact magnetic separatrix, O-point, X-point, or closed-flux volume.

Those quantities will eventually require reconstruction of a poloidal flux function such as $\psi(r, z)$.

4.5. Radial Magnetic Evolution

Figure 2 compares the radial midplane B_z profiles at 2, 3, and 4 μs .

The radial evolution supports the scalar observation that the magnetic configuration does not merely scale uniformly with the imposed field.

Instead, the field structure changes dynamically as the ramp proceeds.

4.6. Density Evolution

The chamber maximum density continues to increase after the original 2 μs baseline endpoint:

$$\rho_{\text{max}}(2 \mu\text{s}) \simeq 50.8,$$

$$\rho_{\text{max}}(3 \mu\text{s}) \simeq 68.3,$$

and

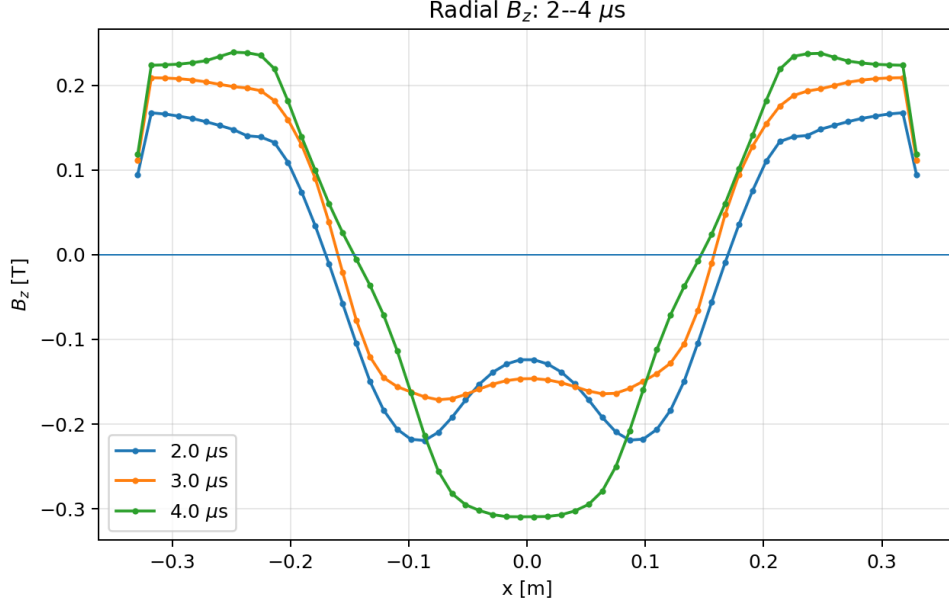


Figure 2: Radial B_z profile through the midplane at $t = 2, 3, 4 \mu s$. The evolving zero crossings and profile amplitudes show that the magnetic response continues to reorganize during the second half of the imposed-field ramp.

$$\rho_{\max}(4 \mu s) \simeq 80.2.$$

The largest sampled chamber maximum occurs slightly before the end of the run,

$$\rho_{\max} \simeq 83.65$$

at approximately

$$t = 3.8 \mu s.$$

The central density does not increase in the same monotonic way. It evolves from approximately

$$\rho_{\text{center}} \simeq 36.2$$

at the strong early compression near $1.7 \mu s$ to

$$12.7$$

at $2 \mu s$, then rises again to approximately

$$24.0$$

at $4 \mu s$.

Therefore, the late increase of $\rho_{\max}$ should not be interpreted as uniform central compression.

A future detailed study should locate the region responsible for the density maximum and relate it to the flux topology and plasma boundary.

4.7. Ion-Temperature Diagnostics

The central ion-temperature diagnostic follows the strong early compression more closely than the chamber maximum.

The largest central value occurs around

$$t = 1.7 \mu\text{s},$$

with

$$T_{i,\text{center}} \simeq 23.2$$

in the units reported by the current WarpX/yt diagnostic pipeline.

By contrast, the pointwise chamber maximum reaches values above

$$3 \times 10^3,$$

with the sampled maximum

$$T_{i,\text{max}} \simeq 3.41 \times 10^3$$

near

$$t = 3.7 \mu\text{s}.$$

This quantity remains unsuitable as a strict macroscopic physical diagnostic.

The earlier regression already demonstrated that pointwise $T_{i,\text{max}}$ is highly sensitive to stochastic PIC sampling.

In addition, the exact physical-unit interpretation of the temperature field should be independently verified before interpreting the large extrema as a quantitative plasma temperature.

4.8. Important Extrema over 0–4 μs

The automatic diagnostic scan found:

Table 2: Selected extrema in the 4000-step formation extension.

Quantity	Extremum	Time
$B_{z,\text{center}}$	$-3.947 \times 10^{-1} \text{ T}$	$1.7 \mu\text{s}$
$B_{z,\text{min}}$	$-4.831 \times 10^{-1} \text{ T}$	$1.7 \mu\text{s}$
$B_{z,\text{max}}$	$+2.600 \times 10^{-1} \text{ T}$	$2.6 \mu\text{s}$
ρ_{max}	8.365×10^1	$3.8 \mu\text{s}$
$T_{i,\text{center}}$	2.322×10^1	$1.7 \mu\text{s}$
$T_{i,\text{max}}$	3.415×10^3	$3.7 \mu\text{s}$

The fact that these extrema occur at different times is another indication that the system has significant transient dynamics and cannot be characterized by a single scalar “compression factor” tied directly to the external ramp.

5. Working Interpretation of the 4 μs Extension

The main observations from the extension are:

- The strongest sampled central magnetic compression occurs near $1.7 \mu\text{s}$, well before completion of the imposed-field ramp.

- A strong relaxation follows before $2 \mu s$.
- The magnetic field subsequently develops a second compression / reorganization stage as the drive approaches its final value.
- Field reversal remains present at $4 \mu s$.
- The chamber maximum density continues increasing until approximately $3.8 \mu s$, while the central density behaves non-monotonically.
- Large pointwise T_i maxima are treated as noisy diagnostics rather than representative bulk plasma temperatures.

These observations are sufficient for the present purpose of the formation extension.

A rigorous formation/compression study would eventually require additional quantities, including

$$\psi(r, z), \quad R_s, \quad L_s, \quad \Phi_{\text{trapped}}, \quad E_{\text{mag}}, \quad E_{\text{kin}},$$

as well as particle and energy-loss diagnostics.

Those additions remain relevant to the long-term framework, but they are not required before moving to the next short-term research objective.

6. Formation Stage Status

The formation workflow has now passed through the following sequence:

legacy takeover $\rightarrow$ baseline audit $\rightarrow$ Framework v0.1 $\rightarrow$ smoke test

$\rightarrow$ 2000-step regression $\rightarrow$ $4 \mu s$ extension.

The validated 2000-step case remains the primary regression reference.

The 4000-step case is retained as the first controlled scientific extension of that reference.

At this point, additional long formation runs are not the highest-priority use of compute time.

The formation module should instead be revisited when new diagnostics, parameter scans, or experimental questions require it.

7. Next Research Stage: GS/MRR and Tilt Reproduction

The main short-term project objective now moves to the inherited Grad-Shafranov / Modified Rigid Rotor equilibrium workflow and the three-dimensional tilt-instability simulations.

The intended sequence is

GS/MRR equilibrium $\rightarrow$ WarpX initialization $\rightarrow$ legacy tilt reproduction $\rightarrow$ validated tilt diagnostics

before introducing new parameter studies.

The known audit questions to resolve include:

1. Alfvén-time normalization

The inherited material contains apparently inconsistent values for the Alfvén time. The normalization used in reported growth rates must be reconstructed from the actual equilibrium parameters.

2. Ion skin depth and S^*

The current implementation must be checked for consistency between deuterium used in the simulation and the mass used when calculating

$$d_i = \frac{c}{\omega_{pi}}.$$

A proton/deuterium mismatch would modify d_i and therefore S^* .

3. Tilt-mode diagnostic

The inherited approximation relating v_z , j_z , and ρ_{protons} must be audited.

A direct particle bulk-velocity diagnostic may be required to avoid confusing total current with ion flow.

4. Growth-rate extraction

The procedure used to obtain

$$\gamma_{\text{tilt}}$$

and the normalization

$$\gamma_{\text{tilt}} \tau_A$$

must be reconstructed and tested quantitatively.

5. Seeded and unseeded runs

The inherited $n = 1$ tilt seed should be reproduced explicitly and compared against the unseeded case before modifying the equilibrium or stability parameters.

8. Framework Development Direction for Week 4

Framework v0.1 has now demonstrated that it can safely reproduce and extend the formation workflow.

The next framework growth should be driven by the GS/MRR and tilt problem rather than by adding generic infrastructure in advance.

Likely next components are:

- a structured equilibrium configuration;
- provenance for the generated `init.bp` openPMD file;
- explicit equilibrium-derived quantities such as R_0 , Z_0 , V_A , τ_A , d_i , and S^* ;
- tilt-seed configuration as a first-class parameter;
- reproducible extraction of the $n = 1$ mode amplitude;
- growth-rate fitting and normalization;
- comparison tools for seeded/unseeded and field-strength cases.

The same project principle remains in force:

reproduce first, then change physics

No refactor or model change should be mixed with the first GS/MRR tilt reproduction.

9. Week 4 Working Status

As of the end of 2026-09-29:

- the inherited 2000-step formation baseline remains validated;
- Framework v0.1 has completed both smoke and full regression tests;
- the same validated formation model has been extended successfully to 4 μs ;
- the 4000-step job completed in 05:29:56 with exit code 0:0;
- the extended run produced 41 diagnostics, 3 checkpoints, and approximately 18 GB of output;
- the strongest early compression occurs around 1.7 μs ;
- a subsequent relaxation and renewed magnetic compression are observed;
- field reversal persists through the end of the imposed-field ramp;
- formation is now sufficiently characterized for the project to move to GS/MRR and tilt-instability reproduction.

Next: reproduce and audit the inherited GS/MRR tilt workflow
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